MOBILE NODE (MN)

A device (laptop, phone) with software allowing it to roam between networks while maintaining connectivity which is identified by its permanent home address.

Example Devices:

- Smartphone
- Laptop
- Tablet
- Drone/ Live Stream Camera

PRIMARY FUNCTION:

- Roams between networks
- Maintains continuous identity
- Manages mobility operations

Key Characteristics:

- Permanent Home Address: 192.168.1.100
- Identity: Layer 3 mobility at IP level i.e. IP Address Dual Semantics, Decoupling Identity and Location, Session Persistence.
- Responsibility: Initiates registration/deregistration

Home Agent (HA): A router on the Mobile Node's home network that intercepts packets meant for the MN's home address, tunnels them to the Care of Agent, and maintains location info.

ROLE: Home network anchor and proxy

- **Primary Functions:**
    1. **Interception:** Captures packets destined for MN's Home Address
    2. **Tunnelling:** Encapsulates packets to forward to Care of Agent
    3. **Tracking:** Maintains binding table of MN locations

Home Agent Implementation Requirements:

- **Must support:** Proxy ARP, IP tunnelling
- **Optional:** Registration revocation, multiple tunnels

- **Security:** Shared secret with each Mobile Node

Foreign Agent (FA): A router on the visited network that provides routing services, acting as a point of attachment for the MN and forwarding tunnelled packets.
The **Foreign Agent (FA)** acts as a temporary "home base" for a device (Mobile Node) when it leaves its original network.

Core Responsibilities of a Foreign Agent

The FA plays a vital role in ensuring a device stays connected as it moves between different networks. Its main duties include:

- **Point of Attachment:** It provides a physical or wireless link for the Mobile Node to connect to the "visited" (foreign) network.
- **Care-of Address (CoA) Management:** The FA provides the MN with a "Care-of Address." This is a temporary IP address that tells the rest of the internet—and specifically the Home Agent—where the device is currently located.
- **De-encapsulation:** When the Home Agent tunnels packets to the MN, they are "wrapped" in another IP header. The FA receives these, strips away the outer layer (de-encapsulation), and delivers the original data to the MN.
- **Registration:** It acts as a middleman, helping the MN register its new location with the Home Agent.

Here is a short breakdown of how it works in four simple steps:

1. Discovery (The "Hello")

When you enter a new network, the Foreign Agent sends out "Agent Advertisements" (like a lighthouse signal). Your device hears this and realizes, *"I'm not at home anymore; I need to use this Foreign Agent to stay connected."*

2. Registration (The "Check-In")

Your device sends a request to the Foreign Agent. The FA then contacts your **Home Agent** (the router at your original house) to say: *"Hey, this device is with me now. Send its mail to my address for a while."*

3. Tunnelling & Delivery (The "Forwarding")

When someone sends data to your permanent IP address:

1. The **Home Agent** intercepts it.

2. It wraps the data in a new "envelope" addressed to the **Foreign Agent** (this is called **Encapsulation**).
3. The Foreign Agent receives the package, opens the outer envelope (**De-encapsulation**), and hands the original data to your device.

4. Sending Data Back

Usually, your device sends data back to the internet directly through the Foreign Agent's router. The FA ensures the packets are routed correctly even though your return address is from a different network.

Care-of Address (CoA): is a temporary IP address assigned to a mobile device (Mobile Node) when it leaves its home network and connects to a "visited" network. The MN's current location on the foreign network, used as the tunnel's endpoint.

While your **Home Address** is like your permanent residence, the **Care-of Address** is like the address of a hotel you are staying at. It tells your Home Agent where to forward your data so you don't lose connection while moving.

There are two main types of Care-of Addresses:

1. Foreign Agent CoA (FA-CoA)

This is the most common type. The address belongs to the **Foreign Agent (FA)** router in the visited network.

- **How it works:** Multiple mobile devices visiting the same network share this single IP address.
- **The Benefit:** It saves IP addresses (crucial for IPv4) and doesn't require the mobile device to have special software to manage its own temporary IP.

2. Co-located CoA (CCoA)

In this version, the mobile device acquires its own individual IP address from the visited network (usually via **DHCP**).

- **How it works:** The device acts as its own "agent." The tunnel from your home network ends directly at your device, not at a router.
- **The Benefit:** You don't need a Foreign Agent to be present on the visited network. However, it "wastes" a unique IP address for every visiting device.

Why is it needed?

Standard IP routing is "topological." This means if you move to a new city but keep your old IP address, the internet routers will still try to send your data to your old city. The **Care-of Address** provides a "forwarding address" so the internet knows exactly which router to hand your packets to right now.